

Split Complex Lorentzian Dynamics

Part XXX: Conditional Global Gluing, Exact Kernel Fibres, and the Last Open Existence Arrow

From compatible whole-overlap transitions to a global internal torsor over the ordinary frame space, with a possible closure isolated for Part XXXI

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Abstract

Part XXIX brought the internal-transition reconstruction to a sharp logical threshold. The locally defined internal representatives of the ordinary frame transitions were no longer merely known to exist after refinement. Their complete repeated-index defect system was proved strong enough to force identity normalization, inverse normalization, exact triple multiplication, and same-pair descent. The apparent second kernel discrepancy κ was shown not to be independent. Consequently, one and only one global compatibility problem remained: whether the continuous kernel-valued equations can be solved simultaneously so that all triple defects vanish. Task 29 proved that this solvability condition is exactly equivalent to the existence of continuous internal transition maps

$$u_{ij} : U_i \cap U_j \longrightarrow \mathcal{L}$$

on the original overlaps satisfying

$$p(u_{ij}) = g_{ij}, \quad u_{ii} = 1, \quad u_{ji} = u_{ij}^{-1}, \quad u_{jk}u_{ij} = u_{ik}.$$

It deliberately did not prove that such a system exists for the inherited arbitrary topological frame family, and it deliberately stopped before constructing the global object that would result if it did [6, 1].

Part XXX answers the second question completely and conditionally. Its input is not the ordinary frame family alone, but an explicitly assumed Task-29 object `CompatibleContinuousInternalTransitions`. From that input, the local products $U_i \times \mathcal{L}$ are glued by the derived transition convention

$$(i, b, a) \sim (j, b, u_{ij}(b)a).$$

The relation is proved reflexive, symmetric and transitive using exactly the identity, inverse and triple laws inherited from Task 29. The resulting quotient `InternalTotal` is therefore a genuine global carrier, not a suggestive disjoint union. The quotient topology is then analyzed rather than accepted formally: it is proved equal to the topology generated by the local product charts, the chart domains are open, the base projection is continuous, and every local coordinate map

$$\mathcal{I}|_{U_i} \cong U_i \times \mathcal{L}$$

is a homeomorphism. Thus the conditional object is topologically locally trivial in the precise sense reconstructed by the task.

A nontrivial left/right issue is also resolved rather than hidden. Since the transition functions act by left multiplication, the globally well-defined internal-group action must act on the right. Task 30 proves that this right $\mathcal{L}$ -action descends through the quotient, is continuous, preserves the base, is free, and is transitive on every fibre. Equivalently, any two internal points over the same base point differ by a unique group element. The local left-coordinate convention

and the global right action are therefore not competing choices; they are two mathematically distinct roles forced by the gluing law.

The ordinary frame side from Task 27 is then reintroduced without modification. Using exactly the inherited identity $p(u_{ij}) = g_{ij}$, the local map

$$(b, a) \longmapsto (b, p(a) \cdot f_{\text{ref}})$$

is proved compatible with both systems of chart changes and descends to a global continuous map

$$\Pi : \mathcal{I} \longrightarrow \mathcal{F}^+.$$

It lies over the identity of the base and is equivariant with respect to the right actions through $p : \mathcal{L} \rightarrow G_{\text{vis}}$. Equality of two internal points under Π is characterized exactly: it holds if and only if the points differ by a unique element of $K(p)$. Each nonempty fibre of Π is therefore a free transitive kernel torsor and is explicitly equivalent to the kernel. Under the certified two-element-kernel assumption, such a fibre has exactly two points. The one-fibre structure reconstructed in Parts XXIV–XXV is recovered chartwise rather than replaced by a new local model.

Task 30 also proves a controlled presentation-independence statement: compatible transition systems related by chart-indexed kernel gauges produce globally equivariantly homeomorphic glued spaces over both the base and the ordinary frame total space. Conversely, internal transition laws can be extracted from a suitable global object, but the projection identity back to the fixed ordinary transition functions requires an explicit **AdaptedCharts** hypothesis. General classification of all presentations and a formal covering-map theorem are intentionally left open.

The supplied run contains twelve new Lean source files and 2399 new Lean lines. The focused Task-30 build is green and the whole project is green at 8340 jobs with zero errors. The Task-30 layer has zero warnings; all 114 whole-project warnings are inherited. Sixty-two audited endpoints expose no nonstandard axiom beyond the established baseline **propext**, **Classical.choice**, and **Quot.sound**. Eighteen failed builds are preserved with exact diagnosis and repair; none changed a mathematical theorem. A minor audit prose inconsistency says “ten new modules” although the listed and delivered chain contains twelve Lean modules. This has no mathematical effect [2].

The Conservation-of-Difficulty frontier is now unusually narrow. Conditional global construction is no longer an open problem: compatible internal transitions are sufficient to generate a global topological object with local product structure, a free and fibrewise transitive internal action, and a continuous equivariant map to the ordinary frame total space with exact kernel fibres. The principal unresolved mathematical arrow is still the one inherited from Part XXIX:

$$\text{ordinary frame transition system} \implies ? \text{ compatible internal transition system.}$$

Before a conventional identification with a spatial spin structure is made, one small formal hardening is also desirable: the current Task-30 layer does not state surjectivity of Π onto the entire ordinary frame total space and does not prove that Π is a covering map, although the inherited transitivity and surjectivity data make the former appear close. Part XXXI should therefore begin with this narrow hardening and then return immediately to the sole remaining existence problem. If that problem closes positively, the spatial branch may reach its first genuine global closure; if it closes negatively, the reconstructed kernel equations will have located the exact obstruction without having imported its conventional name in advance.

Keywords: global gluing; quotient topology; local triviality; internal transition functions; free group action; kernel torsor; frame total space; conditional construction; spin frontier; formal verification; Conservation of Difficulty.

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1 One open arrow from a global object

The transition from Part XXIX to Part XXX changes the nature of the reconstruction. Through Part XXIX, the central question was whether locally available internal representatives can be made mutually coherent. The output was an exact criterion, not a global geometric object. The criterion is compact:

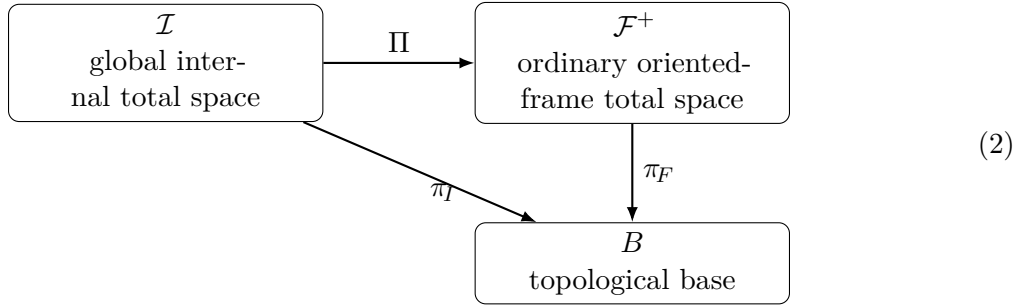
$$p(u_{ij}) = g_{ij}, \quad u_{ii} = 1, \quad u_{ji} = u_{ij}^{-1}, \quad u_{jk}u_{ij} = u_{ik}. \quad (1)$$

The unresolved issue was whether a family satisfying equation (1) exists for the inherited ordinary frame transition system.

A natural temptation would have been to attack that existence question immediately. Task 30 instead performs a logically cleaner experiment: assume the criterion is satisfied and ask whether it actually produces the global object that motivated it. This is an important falsification test. If compatible transition functions did *not* glue to the expected global topology and group action, then the criterion found in Part XXIX would have been insufficient even if solvable. Conversely, if the gluing succeeds, the meaning of the remaining existence question becomes exact.

Methodological principle 1 (Separate existence from construction). A global compatibility predicate should not be interpreted as the existence of the intended global object until a construction theorem has been proved. Conversely, a conditional construction theorem does not prove that its input exists. The two arrows must remain separate in the dependency graph.

Task 30 closes the construction arrow. The resulting diagram is



with a right $\mathcal{L}$ -action upstairs and the inherited right G_{vis} -action downstairs, intertwined by $p : \mathcal{L} \rightarrow G_{\text{vis}}$.

What is new is not merely the familiar shape of equation (2). What matters is its dependency position. The object is obtained only after the ordinary topological frame family, the internal two-to-one group projection, the local representability analysis, the kernel defect system, and the full compatibility criterion have been independently reconstructed. No prepackaged principal bundle or spin bundle is imported to shortcut the gluing.

2 Formal source, scope, and provenance

The supplied Task-30 archive has SHA-256

714e76422de70e5eab644780655a8084feb46209f8baa6fe6251a7f3feefbcce.

The pinned environment is Lean 4.28.0 and Mathlib commit

8f9d9cff6bd728b17a24e163c9402775d9e6a365.

The Task-XXVII, XXVIII and XXIX source layers are byte-for-byte unchanged relative to the preceding archive.

The delivered Task-30 chain contains twelve Lean files:

GlobalGlueBase → InternalGlueRelation → InternalQuotient → InternalTopology →
InternalGroupAction → GlobalFrameProjection → KernelFibres → LocalComparison →
ConditionalGlobalInterface → PresentationIndependence → OrdinaryFrameSideInstance →
Task30.

The audit prose calls this “ten new modules” although its own list and the delivered tree contain twelve. This is a documentation-only mismatch.

Check	Result	Detail
Focused Task-30 build	PASS	lakebuildRequestProject.Null SectorTask30.Task30
Whole-project build	PASS	8340 jobs, 0 errors
Task-30 warnings	0	114 whole-project warnings are inherited
New source	12 Lean files	2399 Lean lines
Axiom audit	62 declarations	only <code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code> , with some definitions requiring subsets
Proof holes	0	no <code>sorry</code> or <code>admit</code>
Prohibited escapes	0	no custom <code>axiom</code> , <code>implemented_by</code> , <code>native_decide</code> , <code>bv_decide</code> , prohibited <code>unsafe</code> , or <code>partial</code>
Preserved failed builds	18	all elaboration, indexing, topology- instance or naming repairs; no theorem weakened

Proof boundary 1 (Verification boundary). *The supplied Task-30 source tree, audit, module dependencies, principal theorem statements, critical proof bodies, negative controls and proof-escape scan were inspected directly for this paper. The successful focused and whole-project builds and the `#print axioms` outputs are reported from the supplied Aristotle artifact. The paper-generation environment did not independently rerun the 8,000-plus-job Lean build.*

Task 30 explicitly does not prove that `CompatibleContinuousInternalTransitions` is inhabited for the inherited family. It does not solve the Task-29 kernel equations, introduce characteristic classes or cohomology, construct connections, add smoothness, or make a physical spin claim. These exclusions are logically central rather than stylistic.

3 Inherited conditional input: the last local-to-global interface

Fix a topological base B and an ordinary open cover $\{U_i\}_{i \in I}$. The ordinary oriented-frame system reconstructed in Task 27 has continuous transition maps

$$g_{ij} : U_i \cap U_j \rightarrow G_{\text{vis}} \quad (3)$$

with the usual identity, inverse and triple-overlap laws.

The internal side supplies a topological group $\mathcal{L}$ and a continuous surjective homomorphism

$$p : \mathcal{L} \rightarrow G_{\text{vis}}, \quad (4)$$

with kernel K . The certified project instance identifies the kernel with the two-element sign subgroup, but the Task-30 gluing layer remains largely kernel-general.

The sole additional hypothesis is an object T of the Task-29 interface `CompatibleContinuousInternalTransitions`. Pointwise, it supplies continuous maps

$$u_{ij} : U_i \cap U_j \rightarrow \mathcal{L} \quad (5)$$

satisfying exactly

$$p(u_{ij}(b)) = g_{ij}(b), \quad (6)$$

$$u_{ii}(b) = 1, \quad (7)$$

$$u_{ji}(b) = u_{ij}(b)^{-1}, \quad (8)$$

$$u_{jk}(b)u_{ij}(b) = u_{ik}(b). \quad (9)$$

These are the only compatibility laws needed by the gluing construction.

Status 1 (Conditional input, not derived). Task 30 assumes equation (5)–equation (9). It neither proves their existence nor weakens the Task-29 equivalence between their existence and simultaneous trivializability of the kernel defect system.

This distinction is what makes Part XXX useful. We can now ask, without prejudging existence, whether these equations are exactly the right sufficient data for the intended global object.

4 Hard target A: the gluing relation is forced

4.1 The tagged local carrier

Before quotienting, Task 30 forms the disjoint union of local products

$$\mathcal{I}_{\text{pre}} = \coprod_i (U_i \times \mathcal{L}). \quad (10)$$

A point is represented by a chart label i , a base point $b \in U_i$, and an internal coordinate $a \in \mathcal{L}$.

The central question is how two chart representatives of one geometric point should be related. The answer is not arbitrary. On the ordinary frame side, Task 27 already established the chart law

$$f_j = g_{ij}(b) \cdot f_i. \quad (11)$$

Part XXIX fixed the internal multiplication convention equation (9). Together with equation (6), these data force the internal chart transformation

$$a_j = u_{ij}(b)a_i. \quad (12)$$

Definition 1 (Internal gluing relation). Two tagged points (i, b, a) and (j, b', a') are related when $b = b'$ and

$$a' = u_{ij}(b)a.$$

4.2 Identity, inverse, and transitivity become quotient consistency

The three normalization equations from Task 29 now acquire a new role. They are exactly the statements required for the relation to be an equivalence relation.

Theorem 1 (Equivalence of the internal gluing relation). *The relation defined by equation (12) is reflexive, symmetric and transitive.*

The proof uses no structure beyond:

- reflexivity: $u_{ii} = 1$;
- symmetry: $u_{ji} = u_{ij}^{-1}$;
- transitivity: $u_{jk}u_{ij} = u_{ik}$.

Thus the normalization work of Part XXIX was not bookkeeping. It is precisely what prevents the global quotient from being ill-defined.

Task 30 also proves a negative control: the relation does not collapse distinct coordinates inside a fixed chart. If two same-chart representatives are related, their internal coordinates agree. Local information therefore survives the quotient.

4.3 The global carrier

The internal total carrier is defined as

$$\mathcal{I} := \mathcal{I}_{\text{pre}} / \sim . \quad (13)$$

The base projection

$$\pi_I : \mathcal{I} \rightarrow B \quad (14)$$

is induced from $(i, b, a) \mapsto b$ and is proved well-defined. Every base point has an internal point because the cover is exhaustive and each local group fibre is nonempty. Hence π_I is surjective.

At this point one has only a quotient set. Task 30 does not mistake this for a topological bundle-like object. The topology is the next hard target.

5 Set-level local coordinates before topology

Fix a chart index i and a point of $\mathcal{I}$ over U_i . A representative may originally be written in some other chart j as (j, b, a_j) . Its coordinate in chart i is transported by the corresponding transition, giving schematically

$$a_i = u_{ji}(b)a_j. \quad (15)$$

Task 30 proves that this expression is independent of the chosen quotient representative. This is the first genuine descent theorem on the quotient itself.

Consequently there is a set-level equivalence

$$\Phi_i : \pi_I^{-1}(U_i) \xrightarrow{\sim} U_i \times \mathcal{L}. \quad (16)$$

Its inverse sends (b, a) to the quotient class of (i, b, a) . Both composite identities are proved explicitly.

Methodological principle 2 (Topology comes after the local bijections). A quotient relation and a family of set-theoretic local product coordinates do not yet determine that the intended local product topology has been obtained. The chart maps must be promoted to homeomorphisms by proof, not by terminology.

This separation mirrors Task 27, where the automatic dependent-sum topology was shown to be too strong and the correct topology had to be reconstructed from an atlas.

6 Hard target B: quotient topology equals chart topology

Task 30 starts from the natural disjoint-union topology on equation (10) and the corresponding quotient topology on equation (13). It then proves that this topology is exactly the topology seen by the local product charts.

Let $q : \mathcal{I}_{\text{pre}} \rightarrow \mathcal{I}$ be the quotient map. The formal endpoints establish:

1. q is a quotient map;
2. the local insertion $\psi_i : U_i \times \mathcal{L} \rightarrow \mathcal{I}$ is open;
3. the image $\pi_I^{-1}(U_i)$ is open;
4. a subset $V \subseteq \mathcal{I}$ is open if and only if every local pullback $\psi_i^{-1}(V)$ is open;
5. the quotient topology is therefore equal to the supremum of the chart-induced topologies;
6. each Φ_i from equation (16) is a homeomorphism;

7. π_I is continuous;
8. the topology is unique among topologies making the same local product charts homeomorphisms in the formal comparison class used by the task.

Theorem 2 (Local topological product structure). *For every i ,*

$$\pi_I^{-1}(U_i) \cong_{\text{Top}} U_i \times \mathcal{L}.$$

The topology on $\mathcal{I}$ is simultaneously the quotient topology of the tagged local carrier and the topology generated by these charts.

This closes a serious possible failure mode. The quotient does not merely have fibres in bijection with $\mathcal{L}$; it is locally topologically the intended product.

6.1 Chart changes

On an overlap, the change of coordinates is

$$(b, a_i) \mapsto (b, u_{ij}(b)a_i). \quad (17)$$

Continuity follows from continuity of u_{ij} and multiplication in the topological group. Task 30 additionally proves that the inverse change is the reversed transition and that compositions satisfy the exact triple law. Thus the topology is not imposed independently of the transition system: it is generated by precisely the same compatibility equations that defined the quotient.

7 The left/right asymmetry is structural

The next issue is easy to mishandle conceptually. Local transitions act on internal coordinates by left multiplication, equation (12). Suppose one attempted to define the global group action also by left multiplication, $a \mapsto ha$. Then a change of chart would compare

$$h(u_{ij}a) \quad \text{with} \quad u_{ij}(ha),$$

which need not agree because $\mathcal{L}$ is not assumed commutative.

Right multiplication, by contrast, commutes automatically with the left transition law:

$$(u_{ij}a)h = u_{ij}(ah). \quad (18)$$

Therefore the globally descending action is forced to be

$$[i, b, a] \cdot h = [i, b, ah]. \quad (19)$$

Theorem 3 (Global internal right action). *The operation equation (19) is well-defined on $\mathcal{I}$, continuous, satisfies the right action laws, and preserves the base projection.*

The formal development explicitly records that there is no contradiction with the earlier one-fibre left presentation. The left action describes coordinate change and model-frame generation; the right action describes global fibre transport after gluing. Their roles are different.

Methodological caveat 1 (Do not erase side information). *Replacing the global right action by a left action merely to match a familiar convention would destroy the proof that the action descends through the chosen gluing relation. Any later conventional comparison must translate conventions rather than silently rewrite them.*

8 Every internal fibre is a torsor

For $b \in B$, define the fibre

$$\mathcal{I}_b = \pi_I^{-1}(b).$$

Any chart containing b identifies this fibre with $\mathcal{L}$, but Task 30 correctly proves that the identification depends on the chart. Changing charts left-translates the coordinate by $u_{ij}(b)$.

The chart dependence disappears at the torsor level.

Theorem 4 (Unique transport in one fibre). *For any $x, y \in \mathcal{I}$ with $\pi_I(x) = \pi_I(y)$, there exists a unique $h \in \mathcal{L}$ such that*

$$x \cdot h = y.$$

Corollary 1 (Fibrewise freeness and transitivity). *The global right $\mathcal{L}$ -action is free and transitive on every fibre of π_I .*

Thus the conditional object has acquired the central global property anticipated by the one-fibre reconstruction: every base fibre is not merely equivalent to the group after choosing a chart; intrinsically it is a free transitive $\mathcal{L}$ -set.

This is a major Conservation-of-Difficulty closure. The local torsor structure from Part XXV survives variation over the base and the global quotient without requiring a global reference frame.

9 Hard target C: descent to the ordinary frame total space

The ordinary topological frame total space $\mathcal{F}^+$ from Task 27 is frozen as inherited data. Its local chart coordinates transform by

$$f_j = g_{ij}(b) \cdot f_i. \tag{20}$$

Let f_{ref} denote the inherited model reference frame. On one internal chart, define

$$(b, a) \longmapsto (b, p(a) \cdot f_{\text{ref}}). \tag{21}$$

The only inherited transition identity used to show that equation (21) descends through the quotient is

$$p(u_{ij}(b)) = g_{ij}(b).$$

Indeed,

$$p(u_{ij}a) \cdot f_{\text{ref}} = p(u_{ij})p(a) \cdot f_{\text{ref}} \tag{22}$$

$$= g_{ij} \cdot (p(a) \cdot f_{\text{ref}}), \tag{23}$$

which is exactly the ordinary frame change equation (20).

Theorem 5 (Global frame projection). *The local maps equation (21) descend to a well-defined continuous map*

$$\Pi : \mathcal{I} \rightarrow \mathcal{F}^+$$

satisfying

$$\pi_F \circ \Pi = \pi_I.$$

Continuity is proved chartwise using the already established product homeomorphisms. No opaque global quotient argument is needed to hide local compatibility.

9.1 Equivariance

The ordinary frame side possesses an inherited right action of G_{vis} . Task 30 proves

$$\Pi(x \cdot a) = \Pi(x) \cdot p(a). \quad (24)$$

This is the global form of the relation between the internal and visible groups.

Status 2 (Global diagram closed conditionally). Given compatible internal transitions, the base projection, internal right action, ordinary frame projection, and group homomorphism form a commuting equivariant system. No local reference choice remains in the statement of equation (24).

10 Exact kernel fibres

The most important classification theorem of the Task-30 global projection is stronger than a mere upper bound on fibre size.

Theorem 6 (Equality downstairs iff kernel transport upstairs). *For $x, y \in \mathcal{I}$,*

$$\Pi(x) = \Pi(y)$$

if and only if there exists a unique $k \in K(p)$ such that

$$x \cdot k = y.$$

The forward direction uses the fibrewise unique $\mathcal{L}$ -transport from theorem 4. If $y = x \cdot a$ and the ordinary images agree, equivariance plus freeness of the visible action force $p(a) = 1$, hence $a \in K$. The reverse direction follows immediately from equivariance because kernel elements are invisible downstairs.

Task 30 packages this as an explicit equivalence

$$\Pi^{-1}(\Pi(x)) \simeq K(p). \quad (25)$$

Thus the fibres over points in the image of Π are not just “two-valued in spirit”. They are exactly the kernel torsors of the internal projection.

Corollary 2 (Two-point fibres for a two-element kernel). *If $\text{card}(K(p)) = 2$, then*

$$\text{card}(\Pi^{-1}(\Pi(x))) = 2.$$

For the certified project instance, previous tasks identify the kernel with $\{\pm 1\}$. Task 30 keeps the cardinality theorem conditional on the explicit kernel-cardinality hypothesis rather than silently specializing the general layer.

Proof boundary 2 (Two-point fibres are not yet a covering theorem). *The result corollary 2 classifies set-theoretic fibres over points in the image. Task 30 does not prove that Π is locally a two-sheeted covering map over $\mathcal{F}^+$. This distinction is preserved explicitly in the audit.*

11 The one-fibre structure survives globalization

Parts XXIV and XXV reconstructed the lifted-frame torsor over one oriented metric frame fibre. It would be a serious failure if Task 30’s quotient produced a different local object.

Task 30 therefore restricts the global construction to a base fibre and compares it in any containing chart with the frozen model. It proves:

- the internal fibre is identified with $\mathcal{L}$;

- the ordinary frame fibre is identified with the inherited model-frame space;
- the global right action corresponds to right multiplication in internal coordinates;
- the global projection is locally $a \mapsto p(a) \cdot f_{\text{ref}}$;
- changing charts modifies the internal coordinate by the inherited left transition and the ordinary coordinate by its visible projection;
- the kernel comparison is preserved.

Methodological principle 3 (Globalization must reproduce, not redefine, the local model). A successful gluing theorem should return the previously reconstructed one-fibre structure when restricted to one fibre. Otherwise the global object would not be a globalization of the established local data but a replacement for them.

This local round trip closes positively.

12 Presentation independence: proved exactly where justified

Compatible transition systems are not unique as presentations. Task 30 introduces a controlled notion **InternalTransitionGauge**: chart-indexed kernel-valued changes relating two compatible systems.

For systems related in this precise way, the task constructs a homeomorphism between the corresponding global quotient spaces and proves that it is

- over the same base;
- equivariant under the global $\mathcal{L}$ -actions;
- compatible with the maps to the same ordinary frame total space.

Therefore a gauge change of the certified kind changes the local presentation but not the global object up to the natural equivariant homeomorphism.

The theorem is deliberately not inflated into a classification statement.

Proof boundary 3 (Presentation-independence boundary). *Task 30 does not prove that every pair of compatible internal transition systems over the same ordinary transitions are related by a chart-indexed kernel gauge. General classification remains open. Only the implication “gauge related $\Rightarrow$ globally equivariantly homeomorphic” is established.*

This is another example of the project’s general rule: a useful invariance theorem is recorded at its exact proved strength rather than converted into an unproved uniqueness claim.

13 Converse extraction and the role of adapted charts

The forward construction shows

$$\text{compatible transitions} \implies \text{global glued object}.$$

Task 30 also tests part of the converse. From a suitable **GluedInternalFrameSystem** equipped with local charts, one can extract transition maps and prove their

- continuity,
- identity law,

- inverse law,
- triple-overlap law.

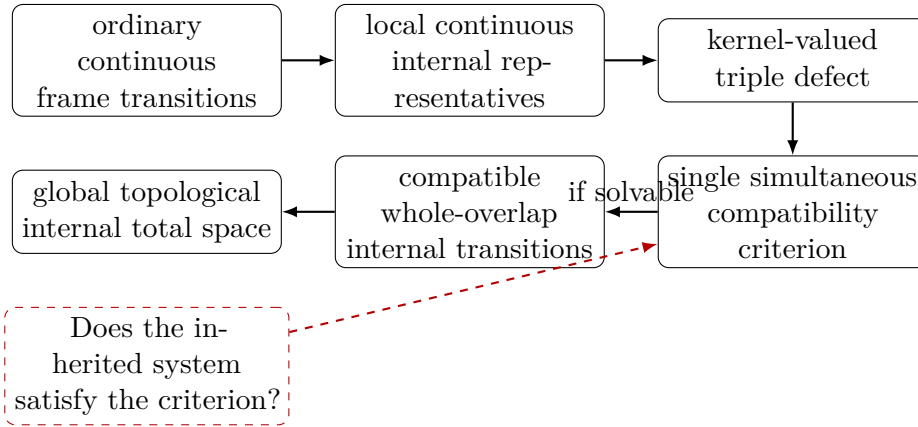
However, to prove that the extracted transitions project to the already fixed ordinary transitions g_{ij} , one needs an additional condition that the internal charts are adapted to the inherited ordinary frame charts. This is packaged as **AdaptedCharts**.

Methodological caveat 2 (An abstract global object does not remember a chosen ordinary atlas automatically). *The existence of local internal trivializations is not by itself enough to identify their projected transition maps with a previously fixed ordinary transition presentation. The relationship between the two atlases is additional comparison data unless derived.*

Thus the converse is correctly recorded as partial, not as an unconditional equivalence of categories or presentations.

14 Conservation of Difficulty: what Task XXX removes

The project has now repeatedly moved the obstruction without erasing it. The progression near the present frontier can be summarized as follows:



Before Task 30, even a positive answer to the existence question would still have left a construction problem: perhaps compatible transitions would fail to give the intended quotient topology, or perhaps the group action would not descend, or perhaps the map to ordinary frames would have larger fibres than the internal kernel. All of those possibilities are now formally eliminated.

Status 3 (Task-30 Conservation-of-Difficulty closure). The global construction problem has closed. The remaining principal difficulty is no longer “how to glue” but “whether the required compatible transition system exists for the inherited family.”

This is exactly the kind of narrowing sought by the reconstruction methodology. The next task can attack one proposition rather than an entire unstructured bundle of local-to-global concerns.

15 A small hardening remains before the conventional spin comparison

Task 30 gets extremely close to the standard geometric shape associated with a twofold lift of an oriented frame structure. Nevertheless, two points should be hardened before any formal identification is stated.

15.1 Surjectivity of the global frame map

The formal Task-30 layer proves:

- $\pi_I : \mathcal{I} \rightarrow B$ is surjective;
- the ordinary visible group acts transitively on each model-frame fibre;
- $p : \mathcal{L} \rightarrow G_{\text{vis}}$ is surjective in the inherited internal projection;
- Π lies over the base;
- Π is equivariant.

These ingredients strongly suggest a short proof that

$$\Pi : \mathcal{I} \rightarrow \mathcal{F}^+ \tag{26}$$

is surjective: choose an internal point over the desired base point, compare its visible image with the desired ordinary frame using fibre transitivity, lift the required visible group element through p , and use equivariance. However, this implication is not exposed as a Task-30 theorem.

Proof boundary 4 (Paper-level diagnostic, not a Task-30 theorem). *Surjectivity of Π appears derivable from already inherited and proved ingredients, but the present paper does not count it as formal Task-30 content. Task 31 should state and prove it explicitly before using a standard global-lift definition that requires coverage of the entire ordinary frame total space.*

15.2 Covering-map strengthening

Likewise, exact two-element fibres do not by themselves prove that Π is a two-sheeted covering map. The project already has local sections and covering information for the one-fibre group projection from earlier tasks, while Task 30 has chart homeomorphisms on both global sides. A narrow next hardening can test whether these combine to local trivializations of Π of the form

$$\Pi^{-1}(V) \cong V \times K$$

for suitable open $V \subseteq \mathcal{F}^+$, and hence to `IsCoveringMap`.

This hardening is technically useful but conceptually secondary to the remaining existence arrow. It should not be allowed to expand into a new broad topology project.

16 Late conventional comparison: why this is almost the spatial spin-bundle diagram

Only after the intrinsic construction is fixed is it appropriate to compare with standard terminology. A conventional spin structure on an oriented Riemannian rank-three bundle is usually described using a principal $\text{Spin}(3)$ -bundle together with an equivariant twofold map to the oriented orthonormal frame bundle, covering the identity on the base and compatible with the standard double covering

$$\text{Spin}(3) \longrightarrow \text{SO}(3)$$

[4, 3, 7].

The project has already reconstructed, in separate formal stages,

- an internal group equivalent to the unit-quaternion/ $SU(2)$ model;
- a visible group equivalent to $SO(3)$;

- the two-to-one group projection with kernel $\{\pm 1\}$;
- the oriented metric frame total space of the varying rank-three family;
- and, conditionally in Task 30, a global locally trivial $\mathcal{L}$ -torsor with an equivariant map to that frame total space whose fibres over image points are exactly kernel orbits.

This is why the present stage is legitimately called the *spin frontier*. But the project still refrains from declaring the conventional identification as a theorem. The reasons are precise:

1. the compatible transition system has not yet been proved to exist for the inherited family;
2. surjectivity of Π has not yet been exposed formally;
3. a global covering-map theorem for Π has not yet been proved;
4. the final standard-interface equivalence has deliberately not been imported as a definition and matched post hoc.

Methodological principle 4 (Identification only after reconstruction). The conventional name should be attached only after the internally reconstructed object has been shown to satisfy the complete target definition. Similarity of diagrams is evidence for the next comparison theorem, not permission to skip it.

Moreover, even a successful spatial $\text{Spin}(3)$ identification would not yet be the full Lorentzian spin structure of spacetime. That later step requires the planned interface between the present spatial branch and the independently reconstructed longitudinal/Lorentz branch.

17 Dependency DAG at the Part-XXX boundary

The mathematical dependency graph at the present stage is

$$\begin{array}{lcl}
\text{ordinary topological frame family} & + & \text{compatible internal transitions [conditional]} \\
\Downarrow & & \\
\text{local products } U_i \times \mathcal{L} & & \\
\Downarrow & & \\
\text{derived gluing relation} & & \\
\Downarrow & & \\
\text{global quotient carrier } \mathcal{I} & & \\
\Downarrow & & \\
\text{quotient topology} = \text{chart-generated topology} & & (27) \\
\Downarrow & & \\
\text{local product homeomorphisms} & & \\
\Downarrow & & \\
\text{free fibrewise transitive right } \mathcal{L}\text{-action} & & \\
\Downarrow & & \\
\Pi : \mathcal{I} \rightarrow \mathcal{F}^+ \text{ continuous and } p\text{-equivariant} & & \\
\Downarrow & & \\
\text{fibres over image points are exactly } K(p)\text{-torsors.} & &
\end{array}$$

The missing incoming edge is

ordinary frame transition system $\implies?$ compatible continuous internal transition system.
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(28)

This edge is not a theorem of Task 30. It is the one principal open arrow carried forward from Part XXIX.

18 Formal decision table

Question	Formal verdict
Compatible internal transitions assumed?	YES
Their existence proved in Task 30?	NO
Gluing relation well-defined and an equivalence relation?	PROVED
Global quotient carrier constructed?	PROVED
Base projection well-defined and surjective?	PROVED
Local set-level charts constructed?	PROVED
Quotient topology equal to chart-generated topology?	PROVED
Local charts homeomorphisms?	PROVED
Base projection continuous?	PROVED
Global $\mathcal{L}$ action descends?	PROVED, as a right action
Global action continuous?	PROVED
Global action free?	PROVED
Global action transitive on each base fibre?	PROVED
Unique group transport in one fibre?	PROVED
Global map to ordinary frame total space?	PROVED
Global map continuous and over the base?	PROVED
Global map equivariant through p ?	PROVED
Fibres over image points exactly kernel orbits?	PROVED
Explicit fibre $\simeq K$ equivalence?	PROVED
Two-point fibre theorem for two-element kernel?	PROVED conditionally on kernel cardinality
Surjectivity of Π onto all of $\mathcal{F}^+$?	NOT EXPOSED AS A TASK-30 THEOREM
Covering-map strengthening?	NOT ATTEMPTED
One-fibre structure recovered locally?	PROVED
Presentation independence?	PARTIAL: gauge-related systems
Converse transition extraction?	PARTIAL: projection law requires adapted charts
Task-29 existence condition solved?	NO
Global obstruction classified?	NO
Conventional spatial spin structure formally identified?	NO
Physical spin derived?	NO

19 Engineering provenance and failed builds

Task 30 preserves eighteen failed builds. None exposed a false mathematical theorem and none required weakening an intended result. The failures are nevertheless retained because they record where the formal dependencies were delicate.

The main classes were:

1. section-variable and binder issues in the frozen ordinary-frame interface;
2. dependent-equality problems while transporting quotient representatives;

3. incorrectly oriented uses of quotient-map continuity lemmas;
4. a genuinely mis-indexed chart-change identity, corrected by recomputing the intended formula rather than weakening it;
5. topology-instance ambiguity in the uniqueness theorem;
6. syntactic mismatch of dependent membership proofs during fibrewise transport;
7. misuse of action notation for a non-instance operation;
8. a wrong namespace for the homeomorphism constructor;
9. an implicit topology argument mismatch in the inherited ordinary-frame instance.

Status 4 (Engineering result). All eighteen repairs preserve the intended mathematical statements. The final focused and whole-project builds are green, and the final documentation follows the green build rather than preceding it.

This provenance is especially important in a quotient-and-topology task. Several failures occurred precisely where definitional equality, dependent membership witnesses, and topology instances can hide mathematical indexing mistakes. The preserved record makes those repairs auditable.

20 What Part XXX has actually accomplished

The strongest valid statement after Task 30 is the following conditional construction theorem.

Theorem 7 (Conditional global construction, informal paper form). *Assume the Task-29 compatible continuous internal transition system exists for the ordinary topological frame family and the frozen internal projection. Then there exists a global topological total space $\mathcal{I}$ over B such that:*

1. $\mathcal{I}$ is locally homeomorphic over each U_i to $U_i \times \mathcal{L}$;
2. $\mathcal{L}$ acts continuously on the right, freely and transitively on each base fibre;
3. there is a continuous map $\Pi : \mathcal{I} \rightarrow \mathcal{F}^+$ over B ;
4. Π is equivariant through $p : \mathcal{L} \rightarrow G_{\text{vis}}$;
5. two internal points have the same ordinary-frame image exactly when they differ by a unique kernel element;
6. hence every fibre of Π over an image point is equivalent to $K(p)$, and has two points when the kernel has cardinality two.

The significance of theorem 7 is that no additional hidden gluing hypothesis appears after Task 29. Once the compatibility equations are solved, the entire global topological construction follows.

21 What remains genuinely open

There are several secondary open questions, but only one principal existence problem.

21.1 The principal existence problem

Task 29 proved

$$\text{simultaneous kernel-defect trivializability} \iff \text{Nonempty}(\text{compatible internal transitions}). \quad (29)$$

Task 30 proves

$$\text{Nonempty}(\text{compatible internal transitions}) \implies \text{Nonempty}(\text{global glued internal object}). \quad (30)$$

What remains is whether the left-hand side of equation (29) holds for the inherited family.

The logical chain is therefore now

$$\boxed{\text{solve the kernel equations} \iff \text{compatible transitions exist} \implies \text{global internal object exists.}} \quad (31)$$

No unknown construction step lies between the final two terms.

21.2 Secondary hardening obligations

Before conventional closure, it is also useful to settle:

1. explicit surjectivity of Π ;
2. if cheap, local two-sheeted covering structure / `IsCoveringMap`;
3. the final comparison with the standard spatial spin-structure definition;
4. only later, broader presentation classification if scientifically useful.

These should remain subordinate to the existence question. The project is now close enough that a broad detour would decrease rather than increase clarity.

22 Outlook: Part XXXI and the possibility of closure

Part XXXI can now be formulated as a narrow closure test rather than another exploratory expansion.

22.1 Stage A: close the tiny Task-30 interface gap

First prove, from already available ingredients, that the global map Π is surjective onto the ordinary frame total space. Then test whether the existing global charts together with the certified local twofold group projection yield local product neighborhoods for Π and hence a covering-map theorem. This stage should be deliberately small.

22.2 Stage B: return immediately to the one remaining global existence equation

The principal target is then the Task-29 condition itself:

$$\exists \varepsilon_{ij} : U_i \cap U_j \rightarrow K \quad \text{continuous, such that} \quad (\varepsilon_{jk} \varepsilon_{ij} \varepsilon_{ik}^{-1}) \delta_{ijk} = 1 \quad (32)$$

for the inherited defect system, with whatever refinement-free equivalent form Task 29 established.

The task should not begin by importing a named characteristic class. It should first determine from the internally derived equations whether the solution exists, fails, or requires an intrinsic global invariant that has not yet been reconstructed. Only after that result is frozen should a late comparison ask whether the discovered criterion agrees with the established obstruction theory [4, 5].

22.3 Possible positive closure

If equation (32) is proved for the relevant inherited family and the small global-map hardening closes, then the present spatial branch will possess:

- the ordinary oriented metric frame family;
- the reconstructed internal double-cover group;
- a globally existing compatible transition system;
- a global locally trivial internal torsor;
- a surjective equivariant twofold map to the ordinary frame total space;
- and, if the comparison theorem closes, the complete conventional spatial spin-structure interface.

That would constitute a genuine closure of this branch, not merely another local resemblance.

22.4 Possible negative closure

A negative theorem would also be informative. If the kernel equations cannot be solved in general, then the reconstruction will have derived the exact global obstruction mechanism from the transition data without assuming its conventional formulation. The next question would then be to classify precisely which ordinary families satisfy the internally reconstructed criterion.

Methodological principle 5 (Closure can be positive or negative). The purpose of Part XXXI is not to force the existence of the desired global lift. It is to close the remaining logical arrow. A proof of existence, a proof of nonexistence in a stated class, or an intrinsic necessary-and-sufficient criterion would each constitute genuine mathematical progress, provided the result is derived from the existing structure rather than inserted as an external answer.

Only after this spatial branch is closed should the project begin the planned formal interface with the independently reconstructed longitudinal/Lorentz branch. That later “marriage” is what will determine whether the spatial spin-like structure and the original split-complex Lorentz structure are two projections of one deeper coordinate-free object.

23 Conclusion

Part XXX reaches the strongest global point yet in the reconstruction. The compatibility criterion of Part XXIX has passed its decisive sufficiency test. Once compatible internal transitions are assumed, there is no further hidden obstruction to constructing the global carrier: the local products glue; the gluing relation is an equivalence relation; the quotient topology is exactly the local-product topology; the base projection is continuous; the internal group acts globally, continuously, freely and transitively on every fibre; and the resulting object maps continuously and equivariantly to the already reconstructed ordinary frame total space. The visible loss of information is exact: equality downstairs is precisely kernel transport upstairs.

The global object is therefore no longer hypothetical in its *construction*. It is hypothetical only in its *existence for the inherited ordinary family*, because the compatible transition input remains conditional. This distinction places the project unusually close to a clean closure:

the construction theorem is closed; the existence theorem is not.

A small interface hardening remains before conventional terminology should be attached: explicit surjectivity of Π , and preferably its local two-sheeted covering structure. Those statements

appear near rather than conceptually new. The substantive frontier is still the one equation system inherited from Part XXIX.

The Conservation of Difficulty has therefore reached a sharp endpoint. The difficulty is no longer spread across local lifts, refinement, topology, quotient consistency, global actions, or kernel fibres. It is concentrated in one question:

Does the internally reconstructed kernel-defect system admit the simultaneous continuous trivialization required to instantiate the now-complete global construction?	(33)
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Part XXXI should answer that question after the narrow surjectivity/covering hardening. If it closes, the spatial branch can finally be compared theorem-for-theorem with the conventional definition that has so far been kept outside the reconstruction firewall. The next task may therefore be not another expansion of the theory, but its first genuine global closure.

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